

AIFS Cross-Attention Mechanism Detail

TimeSeries Tokens ↔ Llama 3-8B Embeddings Fusion

Mathematical Formulation

- Multi-Head Cross-Attention:
- Projection: $Q = X_{\text{text}} W_Q$, $K = V = X'_{\text{climate}} W_{K,V}$
 - Multi-Head: $Q_h = Q W_h^Q$, $K_h = K W_h^K$, $V_h = V W_h^V$
 - Attention: $A_h = \text{softmax}(Q_h K_h^T / \sqrt{d_k})$
 - Output: $O_h = A_h V_h$
 - Concatenate: $O = \text{Concat}(O_1, \dots, O_{32}) W_O$
 - Residual: $Y = \text{LayerNorm}(X_{\text{text}} + O)$

Attention Matrix
 $A: [B, 32, 128, 64]$
Text pos × Climate pos
Temperature-scaled
 $\tau = 0.1$ (learnable)

AIFS Cross-Attention Specifications:

- Input Dimensions:
- Climate: $[B, T, 218] \rightarrow$ projected to $[B, T, d_{\text{llm}}]$
 - Text: $[B, \text{seq_len}, d_{\text{llm}}]$ (native LLM dimension: 768 or 4096)
- Multi-Head Configuration:
- Heads: 32 (for Llama 3-8B) or 12 (for smaller models)
 - Per-head dimension: 128 ($4096 \div 32$) or 64 ($768 \div 12$)
 - Total parameters: ~67M for attention layers
- Attention Mechanism:
- Query: from text embeddings
 - Key/Value: from projected climate embeddings
 - Temperature scaling: learnable $\tau \in [0.01, 1.0]$
 - Dropout: 0.1 during training

Cross-modal
attention matrix

Input Token Representations

AIFS Climate Embeddings
 $X_{\text{climate}} = [B, T, 218]$
B=1, time=2, d_model=218
From AIFS encoder output

LLM Text Tokens
 $X_{\text{text}} = [B, \text{seq_len}, d_{\text{llm}}]$
B=1, seq_len variable, d_llm=768/4096
From climate queries

Dimension Alignment Layer

Climate Projector
 $W_c: 218 \rightarrow d_{\text{llm}}$
Linear(218, 768/4096) + LayerNorm

Projected Climate
 $X'_{\text{climate}} = [B, T, d_{\text{llm}}]$
Aligned with LLM dim

Climate tokens
projected to
LLM dimension

Multi-Head Cross-Attention Computation

32 attention heads
parallel computation

Query Projection
 $Q = X_{\text{text}} \cdot W_Q$
Q: $[B, 128, 4096]$

Key Projection
 $K = X'_{\text{climate}} \cdot W_K$
K: $[B, 64, 4096]$

Value Projection
 $V = X'_{\text{climate}} \cdot W_V$
V: $[B, 64, 4096]$

Multi-Head Split
32 heads of $d_k=128$
Parallel computation

Attention Computation
 $A_h = \text{softmax}(Q_h K_h^T / \sqrt{d_k})$
Per-head attention weights

Output Fusion & Integration

Head Concatenation
 $\text{Concat}(O_1, \dots, O_{32})$
[B, 128, 4096]

Output Projection
 $W_O: 4096 \rightarrow 4096$
Linear + Dropout

Fused Embeddings
 $Y = [B, 128, 4096]$
Text enhanced with climate context
Ready for Llama decoder